

**B.Tech. Degree II Semester Regular and Supplementary/  
I Semester Supplementary Examination May 2017**

CS/EC/IT AS 15-1201 B & CE/EE/ME/SE AS 15-1101 A

**CALCULUS**

(2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

**PART A**

(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Solve  $(3x^2 + 6xy^2)dx + (6x^2y + 4y^3)dy = 0$ .
- (b) Solve  $\sec^2 y \frac{dy}{dx} + 2x \tan y = x^3$ .
- (c) Solve  $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 3x = \sin t$ .
- (d) Test the convergence of the series  $\sum \frac{1}{\sqrt{n} + \sqrt{n+1}}$ .
- (e) Find the Maclaurin series expansion of  $\tan^{-1} x$ .
- (f) If  $x^y + y^x = c$ , find  $\frac{dy}{dx}$ .
- (g) Show that  $U = e^x \sin y$ ,  $V = e^x \cos y$  are functionally independent.
- (h) Find the perimeter of the cardioid  $r = a(1 - \cos \theta)$ .
- (i) Find the area enclosed by the curve given by the equation  $x = a \cos^3 \theta$ ,  $y = b \sin^3 \theta$ .
- (j) Evaluate  $\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} dx dy dz$ .

**PART B**

(4 × 10 = 40)

- II. (a) Solve  $\frac{d^4x}{dt^4} + 4x = 0$ .
- (b) Solve  $\frac{d^2y}{dx^2} - \frac{2dy}{dx} + y = xe^x \sin x$ .
- (c) Solve  $x^2 \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} - 20y = (x+1)^2$ .

**OR**

(P.T.O.)



- III. (a) Solve the simultaneous equation.

$$\frac{dx}{dt} - x + \frac{dy}{dt} = 2t + 1$$

$$2\frac{dx}{dt} + x + \frac{2dy}{dt} = t.$$

- (b) A condenser of capacity  $C$  discharged through inductance  $L$  and resistance  $R$  in series and the charge  $q$  at time  $t$  satisfies the equation

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = 0, \text{ where } L = 1 \text{ henry, } R = 400 \text{ ohms, } C = 16 \times 10^{-6} \text{ farads.}$$

Find  $q$  in terms of  $t$ .

- IV. (a) Find the nature of the series.

$$\frac{1}{2} + \frac{2}{3}x + \left(\frac{3}{4}\right)^2 x^2 + \left(\frac{4}{5}\right)^3 x^3 + \dots$$

- (b) State Raabe's test and test the convergence of

$$1 + \frac{1}{2}x + \frac{1.2}{2.5}x^2 + \frac{1.2.3}{2.5.8}x^3 + \dots$$

OR

- V. (a) Define conditional convergence and absolute convergence of series with examples.

- (b) If  $y = e^{a \sin^{-1} x}$ , prove that  $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 + a^2)y_n = 0$ .

Hence find the value of  $y_n$  when  $x = 0$ .

- VI. (a) Show that if  $u = \sin^{-1} \frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$  then  $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} + 3 \tan u = 0$ .

- (b) If  $u = \frac{y}{z} + \frac{z}{x} + \frac{x}{y}$ , show that  $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$ .

- (c) The diameter and height of a right circular cylinder are measured to be 5 and 8 inches respectively, if each of these dimensions are in error 0.1 inch. Find the relative percentage error in volume of the cylinder.

OR

- VII. (a) If  $x = r \sin \theta \cos \phi$ ,  $y = r \sin \theta \sin \phi$ ,  $z = r \cos \theta$ . Show that

$$\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = r^2 \sin \theta.$$

- (b) A rectangular box open at the top is to have a volume 32 c.c. Find the dimensions of the box requiring least material for its construction.

- VIII. (a) Find the area of the surface of revolution formed by revolving the loop of the curve  $9ay^2 = x(3a-x)^2$  about  $X$ -axis

- (b) Evaluate the integral by changing the order of integration  $\int_0^a \int_{x/2}^{2a-x} xy dy dx$ .

OR

- IX. (a) Calculate  $\iint r^3 dr d\theta$  over the area included between the circles  $r = 2 \sin \theta$  and  $r = 4 \sin \theta$ .

- (b) Evaluate  $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dz dy dx}{\sqrt{1-x^2-y^2-z^2}}$ .

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**B.Tech. (Part-Time) Degree I Semester Supplementary Examination**  
**April 2019**

**CE/ME/CH AS 17P 1101 LINEAR ALGEBRA AND CALCULUS**  
(2017 Scheme)

Time: 3 Hours

Maximum Marks: 60

**PART A**  
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Define the characteristic equation of a square matrix. Give an example of a characteristic equation of a  $2 \times 2$  square matrix.
- (b) Define the homogeneous function in two variables.
- (c) Find the error in the area of a circle of radius  $r = 5$  cm with an error  $\delta r = 0.2$  cm in the radius.
- (d) Solve  $(D^2 + 3D + 2)y = 0$
- (e) Find the area enclosed between the parabola  $y = x^2 - 5x + 6$  and the  $x$ -axis.
- (f) Define the normal form of a matrix.
- (g) State Cayley Hamilton theorem. Give an example.
- (h) Solve  $(y + x^2)dx + (x + y^2)dy = 0$ .
- (i) Define the linear differential equation with constant coefficients.
- (j) If  $x = uv$  and  $y = \frac{u}{v}$ , find  $\frac{\partial(x, y)}{\partial(u, v)}$ .



**PART B**

(4 × 10 = 40)

- II. (a) Find the Eigen values and Eigen vectors of the matrix  $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ .
- (b) Solve the system
- $$\begin{aligned} x + y + z + w &= 0 \\ x + y + z - w &= 4 \\ x + y - z + w &= -4 \\ x - y + z + w &= 2 \end{aligned}$$

OR

- III. (a) Verify Cayley Hamilton theorem for the matrix  $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ .
- (b) Is the matrix  $\begin{bmatrix} 3 & 4 \\ 5 & 2 \end{bmatrix}$  diagonalizable?

(P.T.O)

- IV. (a) If  $u = \sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$ , prove that  $\sin u$  is a homogeneous function and that  $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}\tan u$ .

- (b) Find all the first and second partial derivatives of  $f(x, y) = \sqrt{x^2 + y^2}$ .

OR

- V. (a) If  $x = u(1-v)$  and  $y = uv$  prove that  $JJ' = 1$ .  
 (b) Examine the following function for extreme values  
 $f(x, y) = xy - x^2 - y^2 - 2x - 2y + 4$

- VI. (a) Solve  $(D^2 + 4D + 5)y = x^2$ .

- (b) Solve  $(D^2 + 3D + 2)y = \sin 2x$ .

OR

- VII. (a) Solve  $\frac{dy}{dx} = \frac{\sin y + y \sin x}{\cos x - x \cos y}$ .

- (b) Solve  $\frac{dx}{dt} + y = \sin t$ ;  $\frac{dy}{dt} + x = \cos x$ , given that  $x = 2$ ,  $y = 0$  when  $t = 0$ .

- VIII. (a) Find the area enclosed between the parabola  $x^2 = 4y$  and  $y^2 = 4x$ .

- (b) Solve  $\int_0^3 \int_1^2 x(x+y) dy dx$ .

OR

- IX. (a) Solve  $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$ .

- (b) Find the volume formed by the revolution of loop of the curve  $y^2(a+x) = x^2(3a-x)$  about the  $x$ -axis.

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***B.Tech. Degree I Semester Regular and Supplementary /  
II Semester Supplementary Examination November 2018***

**CE/EE/ME/SE AS 15 - 1101 A & CS/EC/IT AS 15 - 1201 B CALCULUS  
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

**PART A  
(Answer ALL questions)**

(10 × 2 = 20)

- I.
- Solve  $xy(1+xy^2)\frac{dy}{dx}=1$ .
  - Solve  $(5x^4+3x^2y^2-2xy^3)dx+(2x^3y-3x^2y^2-5y^4)dy=0$ .
  - Solve  $(D^3+1)y=0$ .
  - Test the convergence of the series  
 $1+\frac{1}{4^{\frac{1}{2}}}+\frac{1}{9^{\frac{1}{2}}}+\frac{1}{16^{\frac{1}{2}}}+\dots$
  - Test whether the series is absolutely convergent or conditionally convergent  
 $1-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+\dots$
  - Expand  $\cos x$  about  $x=\pi/2$  using Taylor's series.
  - If  $u=\tan^{-1}\frac{x^3+y^3}{x-y}$  prove that  $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\sin 2u$ .
  - Show that  $u=\frac{x+y}{x-y}$ ,  $v=\frac{xy}{(x-y)^2}$  are functionally dependent.
  - Find the area of the curve  $r=2a\cos\theta$ .
  - Change the order of integration in the integral and sketch the region  $\int_0^a \int_{x^2/a}^{2a-x} f(x,y)dx dy$ .



**PART B**

(4 × 10 = 40)

- II.
- Solve  $\frac{d^2y}{dx^2}+y=e^{2x}+x\cos x$ .
  - Solve  $x^2\frac{d^2y}{dx^2}+2x\frac{dy}{dx}-20y=x^4$ .

OR

- III.
- Solve  $\frac{dx}{dt}=2y-1$ ;  $\frac{dy}{dt}=2x+1$
  - A voltage  $Ee^{-at}$  is applied at  $t=0$  at a circuit containing inductance  $L$  and resistance  $R$ . Show that the current at any time  $t$  is  $\frac{E}{R-aL}(e^{-at}-e^{-\frac{R}{L}t})$  assuming that  $i=0$  at  $t=0$ .

(P.T.O.)

- IV. (a) Determine the nature of series  $\sum \left[ \left( \frac{n+1}{n} \right)^{n+1} - \left( \frac{n+1}{n} \right) \right]^n$ .
- (b) If  $y = \cos(m \sin^{-1} x)$ , prove that  $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 - m^2)y_n = 0$ .

OR

- V. (a) Show that series  $\sum_{n=0}^{\infty} \frac{(n!)^2 x^n}{(2n)!}$  is convergent for  $x < 4$  and divergent for  $x \geq 4$ .
- (b) Expand  $\sin x$  in powers of  $(x - \pi/2)$ . Hence find the value of  $\sin 91^\circ$ .

- VI (a) If  $u$  is a homogeneous function of degree  $n$  in  $x$  and  $y$  show that  $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$ .
- (b) Find the maximum and minimum values of  $f(x, y) = x^3 + y^3 + 3xy$ .

OR

- VII. (a) If  $u = f(r, s, t)$  and  $r = \frac{x}{y}$ ,  $s = \frac{y}{z}$ ,  $t = \frac{z}{x}$ , prove that  $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$ .
- (b) Find the shortest and longest distance from the point  $(1, 2, -1)$  to the sphere  $x^2 + y^2 + z^2 = 24$ .

- VIII. (a) Find the volume of the solid generated by revolving  $x^{2/3} + y^{2/3} = a^{2/3}$  about the  $x$ -axis.
- (b) Evaluate  $\iint_R \frac{e^{-y}}{y} dx dy$  by choosing order of integration suitably given that, region 'R' is bounded by the lines  $x = 0$ ,  $x = y$ , and  $y = \infty$ .

OR

- IX. (a) Find the area of the crescent bounded by  $r = \sqrt{3}$  and  $r = 2 \cos \theta$ .
- (b) Evaluate  $\int_0^1 \int_{y^2}^1 \int_0^{1-x} x dz dx dy$ .

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BTS-II(R)&amp;I(S)-05.16-0769

Reg. No.

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**B.Tech. Degree II Semester Regular and I Semester Supplementary  
Examination May 2016**

**IT/CS/EC/CE/EE/ME/SE 1101 A / 1201 B CALCULUS  
(2015 Scheme)**

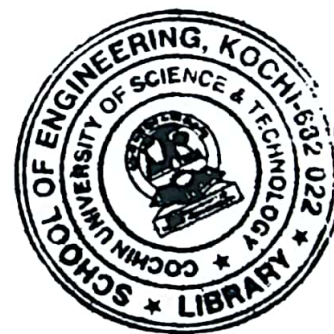
Time : 3 Hours

Maximum Marks : 60

**PART A  
(Answer ALL questions)**

(10 × 2 = 20)

- I. (a) Solve  $\frac{dy}{dx} = e^{2x-y} + x^3 e^{-y}$ .
- (b) Solve  $\frac{d^2 y}{dx^2} + 25y = \cos 4x$ .
- (c) Solve  $x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y = e^x$ .
- (d) Test the convergence of the series  $\sum_{n=0}^{\infty} \left(\frac{1}{2^n}\right)$ .
- (e) Explain the Cauchy's root test and give one example.
- (f) Obtain the Maclaurin series expansion of  $f(x) = \sin x$ .
- (g) Find  $\frac{\partial^3 u}{\partial x \partial y \partial z}$ , if  $u = e^{x^2+y^2+z^2}$ .
- (h) Find the Jacobian  $\frac{\partial(u,v)}{\partial(x,y)}$  of  $u = x \sin y$ ;  $v = y \sin x$ .
- (i) Find the area bounded by the parabolic arc  $\sqrt{x} + \sqrt{y} = \sqrt{c}$ ,  $c > 0$  and the coordinate axes.
- (j) Evaluate  $\int_0^a \int_0^{\sqrt{a^2-y^2}} (x^2 + y^2) dx dy$ .



**PART B**

(4 × 10 = 40)

- II. (a) Solve  $\frac{dy}{dx} = \frac{y-x}{y-x+2}$ .
- (b) The slope at any points (x,y) of a curve is  $1 + \frac{y}{x}$ . If the curve passes through (1,1) find the equation of the curve.
- (c) Solve  $4y^2 p^2 + 2pxy(3+1) + 3x^3 = 0$ , where  $p = \frac{dy}{dx}$ .

OR

(P.T.O.)

III. (a) Solve  $\frac{dy}{dx} + 2y = e^x(3\sin 2x + 2\cos 2x)$ .

(b) Solve  $4x^2 \frac{d^2y}{dx^2} - 4x \frac{dy}{dx} + 3y = 0$ .

(c) Solve the simultaneous equations

$$\frac{dy}{dt} + 5y - 2x = t$$

$$\frac{dx}{dt} + 2y + x = 0$$

IV. (a) Explain Raabe's test for testing the convergence of an infinite series and using it test the convergence of the series  $\sum_{n=1}^{\infty} \left[ \frac{4.7 \dots (3n+1)}{1.2 \dots n} \right] x^n$ .

(b) Test for the uniform convergence of the series

$$\frac{\sin x}{1^2} + \frac{\sin 2x}{2^2} + \frac{\sin 3x}{3^2} + \frac{\sin 4x}{4^2} + \dots$$

OR

V. (a) Discuss the convergence of the series

$$1 + \frac{x}{2} + \frac{x^2}{5} + \dots + \frac{x^n}{n^2 + 1} + \dots$$

(b) Obtain power series solution in powers of  $x$  for the following differential

$$\text{equation: } (x^2 - 1) \frac{d^2y}{dx^2} + 3x \frac{dy}{dx} + xy = 0, \quad y(0) = 4 \text{ and } y'(0) = 6.$$

VI (a) If  $u = x^3 y^2 \sin^{-1}(y/x)$  show that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 5u.$$

(b) Show that the rectangular solid of maximum volume that can be inscribed in a given sphere is a cube.

OR

VII. (a) If  $V = u^2 v$  and  $u = e^{x^2 - y^2}$ ,  $v = \sin(xy^2)$  find  $\frac{dv}{dx}$  and  $\frac{dv}{dy}$ .

(b) Given  $u = 2\alpha xy$ ,  $v = a(x^2 - y^2)$  where  $x = r \cos \theta$ ,  $y = r \sin \theta$ , find  $\frac{\partial(u, v)}{\partial(r, \theta)}$ .

VIII. (a) Find the total area between the cubic  $y = 2x^3 - 3x^2 - 12x$ ,  $x$ -axis and its maximum and minimum ordinates.

(b) Evaluate  $\iint_R (x+y)^2 dx dy$ , where  $R$  is the parallelogram in the  $xy$ -plane with vertices  $(1,0)$ ,  $(3,1)$ ,  $(2,2)$ ,  $(0,1)$  using the transformation  $u = x + y$  and  $v = x - 2y$ .

OR

IX. (a) What is the volume generated by revolving the area enclosed by the loop of the curve  $y^4 = x(4-x)$  about  $x$ -axis?

(b) Find the mass, centroid of the tetrahedron bounded by the coordinate planes and the plane  $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ . Given that the substance has constant density  $\rho$ .